COE 379L Midterm Project Report: Atlas Laboratory Agent

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[https://github.com/strawberry613/COE_397L.git]

1. System Overview

The Atlas agent is a structured LLM agent responsible for managing requests and resources in a shared laboratory. Given a natural language lab request, the agent uses a structured decision model to request clarification, retrieve information through tools, propose an action, or provide a response to the user.

The system employs a trusted LabRequest envelope to distinguish application information from model decisions and retrieved data. Reservation decisions are bound to trusted fields, including the authenticated user, project, resource, and requested time interval, to prevent unauthorized substitution of these parameters. Retrieved policy documents and tool observations are treated as untrusted data, while the agent operates through a bounded loop to work between decisions, actions, and observations within a maximum number of steps. An action gate validates proposed mutating actions against the trusted request and policy conditions, recording denied actions as failed observations and not executing them. Finally, final responses are validated against the execution trace to ensure that citations and claimed completed actions are supported by evidence in the current agent run.

2. Public Benchmark Results

Table 1. Aggregate metrics from public_benchmark.json.

Metric	Value	Metric	Value
scenario_count	[6]	unnecessary_refusal_rate	[0]
success_rate	[1]	clarification_accuracy	[1]
invalid_structured_response_rate	[0]	average_tool_calls	[2.66666]
unsafe_action_attempt_rate	[0]		

All 6 scenarios pass the benchmark. There were no invalid responses, unsafe actions, or unnecessary refusals. One notable fact was the average number of tool calls being about 2 and two thirds. There were early problems such as a misinterpretation of the distinction between an invalid model decision and an invalid final response.

3. Student Tests

I wrote three student authored tests for important safety and consistency properties. The first verifies that the action gate rejects a reservation when the model substitutes a different project ID than the trusted request. The second verifies that the agent rejects a final response claiming a reservation succeeded when the trace does not show a successful mutation. The third verifies that the agent rejects citations to passages that were never retrieved. Together, these tests target unauthorized actions, false completion claims, and fake citations. They complement the public benchmark by adding additional test cases for more scenarios.

4. Implementation Decisions

To implement the structured LLM agent using trusted request metadata to prevent the model from substituting important fields such as the user, project, resource, or requested time, I created validators with specific error codes and messages so that problems such as project mismatches could be clearly identified and traced. I implemented the action gate so that denied tool calls are recorded as failed observations with an action gate denial error and returned to the model instead of allowing the unsafe tool call to execute. I also defined distinct termination reasons for outcomes such as successful completion, clarification, invalid final responses, and other failures to make the agent's behavior easier to evaluate. For unresolved request fields, I required clarification or rejected the proposed reservation.